

RetailHub SQL Project Diagrams

Diagram Documentation

MySQL • SQL • Database Design • Data Analytics

RetailHub is a complete SQL Data Analytics project that simulates the database of a real-world e-commerce company. The project demonstrates how relational databases are designed, populated, and analyzed using MySQL.

The system manages customers, products, suppliers, warehouses, inventory, orders, payments, shipments, returns, coupons, wishlists, and customer support. Using SQL queries and database objects, it generates business insights that help management make data-driven decisions.

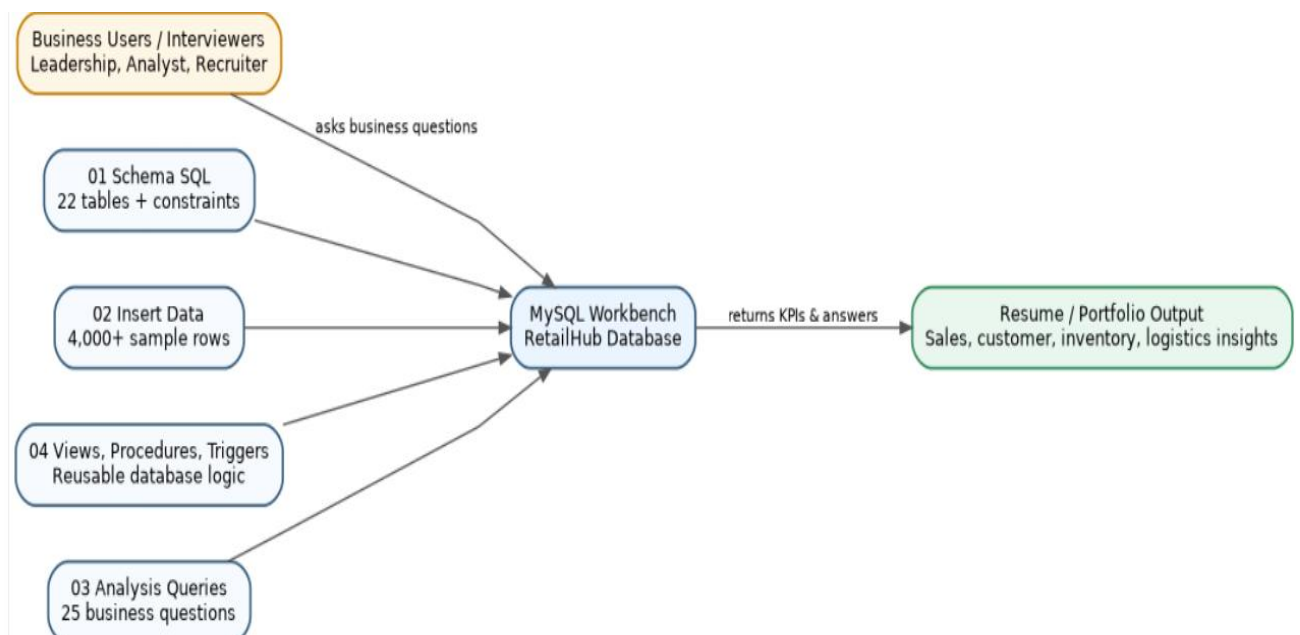
This project showcases practical SQL skills including database design, normalization, joins, subqueries, CTEs, window functions, views, stored procedures, functions, and triggers.

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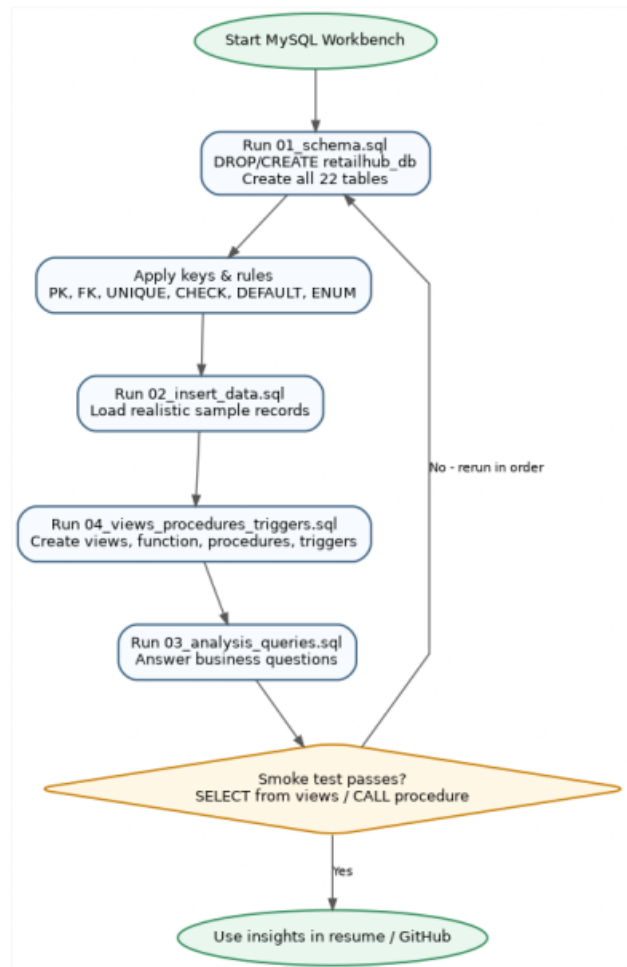
❖ Project Architecture Diagram

This diagram shows how the SQL files work together to create the complete RetailHub project. It is useful in a resume or viva because it proves this is not only a few SELECT queries, but an end-to-end analytics system. The schema creates the database, sample data fills it, database logic adds reusable features, and analysis queries generate business insights. It helps the viewer understand the full project in one glance.



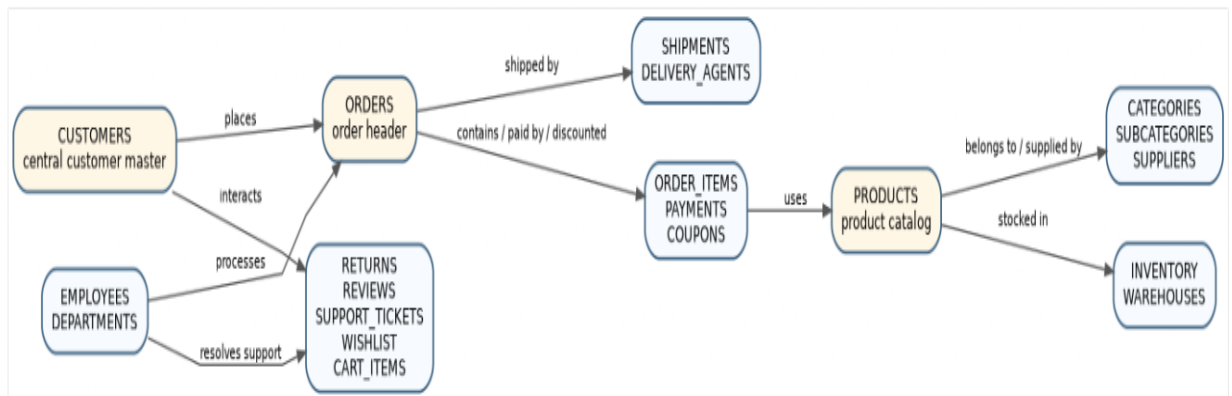
❖ Database Creation Flow Diagram

This flow explains the correct execution order of the project files in MySQL Workbench. It is used to avoid confusion during setup because schema must be created before inserting data, and views/procedures/triggers should be created before final analysis. This diagram is also helpful during project demonstration because it shows a clean, repeatable workflow. The smoke test step confirms that the database is ready for analysis.



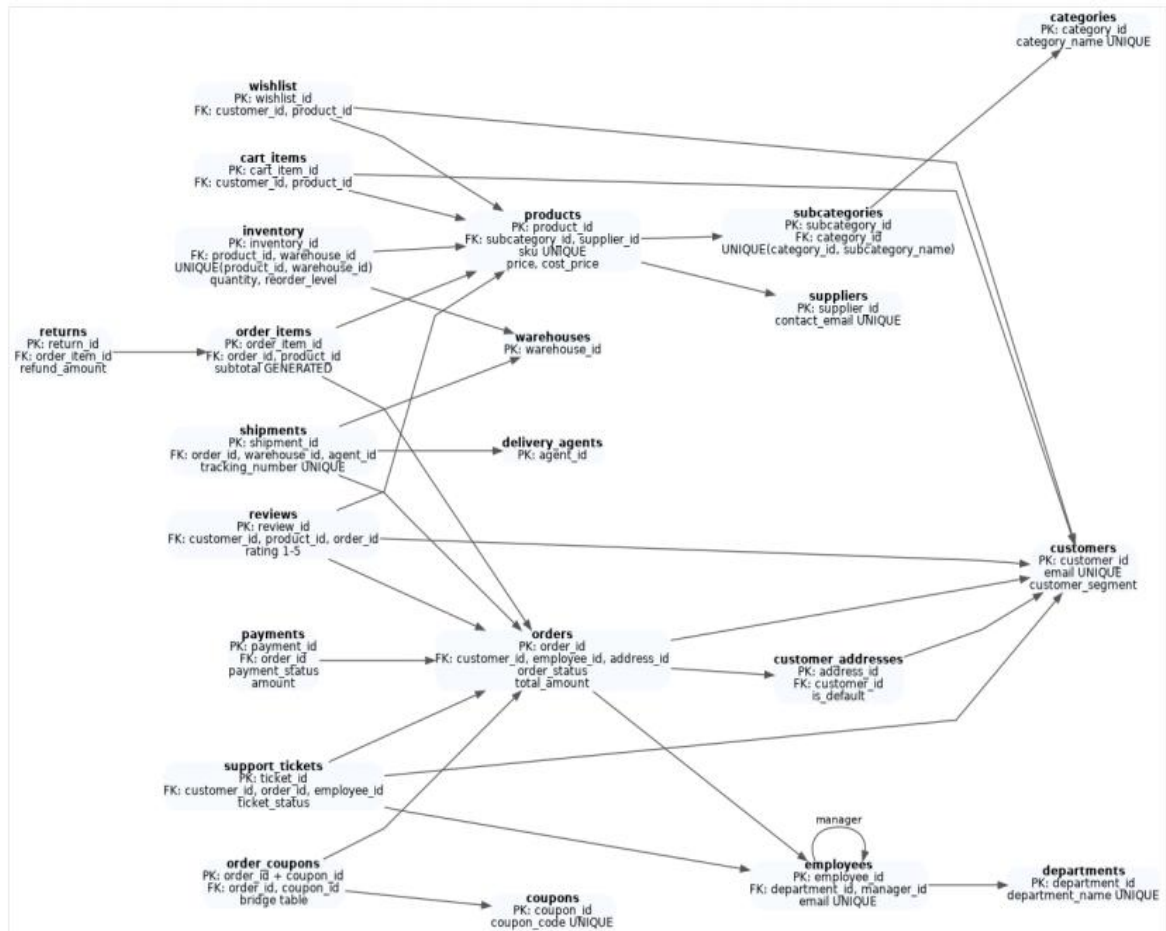
❖ ER Diagram

The ER diagram shows the main entities and how they are related in the e-commerce business. It is used to explain the database design before going into table-level detail. Customers place orders, orders contain products, payments and shipments complete the sale, and reviews/returns/support capture post-sales activity. This is the best diagram to put in a resume or GitHub README for quick understanding



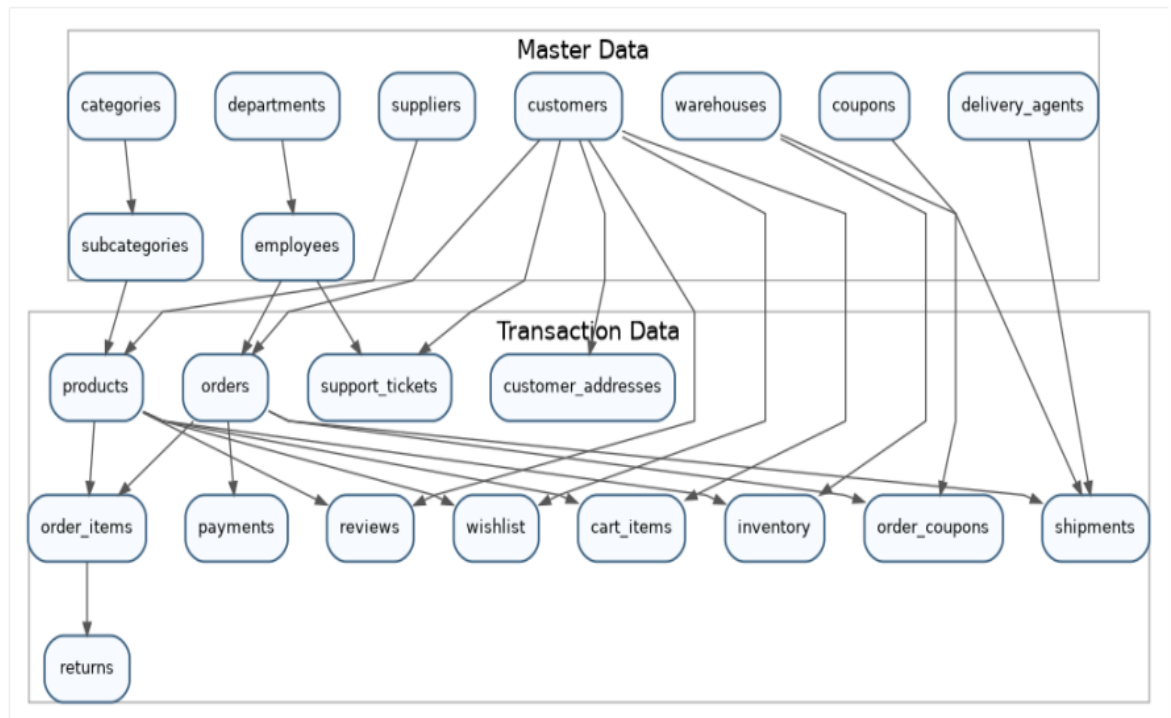
❖ ER Diagram with Tables and Keys

This diagram shows the important primary keys, foreign keys, bridge tables, and unique constraints. It is used to explain how the 22 tables are connected and how referential integrity is maintained. The order_coupons table is shown as a bridge table because many orders can use coupons and each coupon can be used in many orders. This diagram is useful for interview questions on normalization, joins, PK, FK, and composite keys.



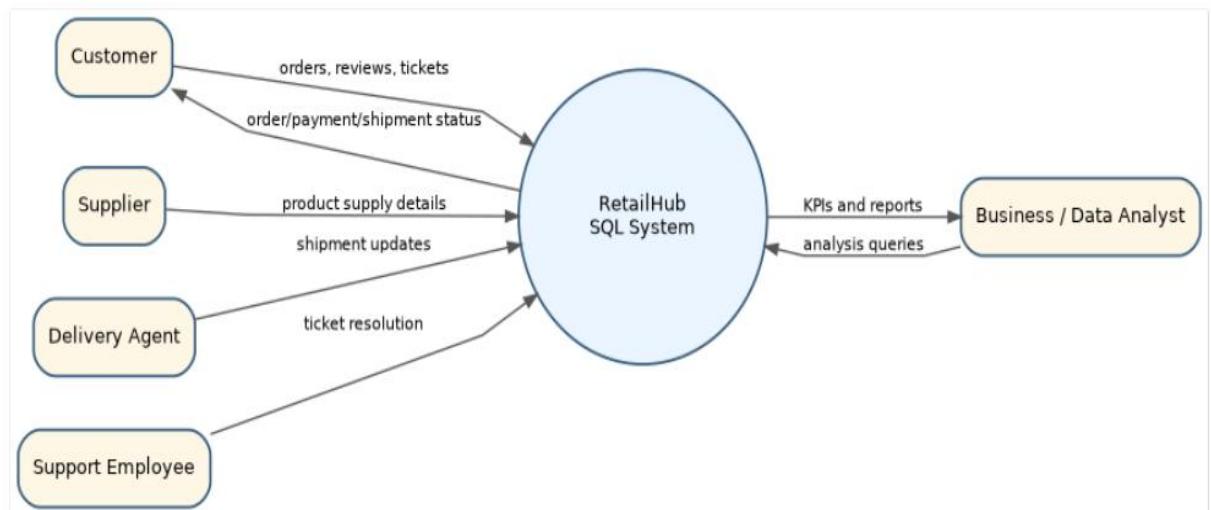
❖ Table Module Map

This diagram groups the 22 tables into master data and transaction data. It is used to make the large schema easier to understand by separating stable reference tables from daily business activity tables. Master tables store entities like customers, employees, suppliers, products, and warehouses. Transaction tables record activity such as orders, payments, shipments, returns, reviews, carts, and support tickets.



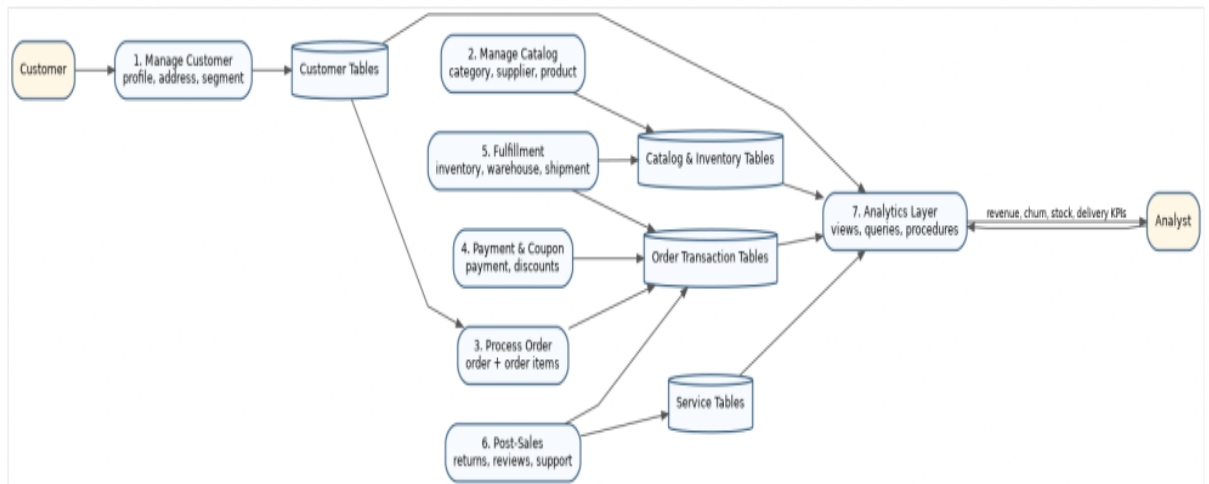
❖ Data Flow Diagram - Level 0 Context

This context DFD shows RetailHub as one complete system and shows the external users that exchange data with it. It is used to explain the project from a business-process point of view rather than a table point of view. Customers, suppliers, delivery agents, support employees, and analysts all interact with the same database system. The diagram shows the system as a single source of truth for operational and analytical data.



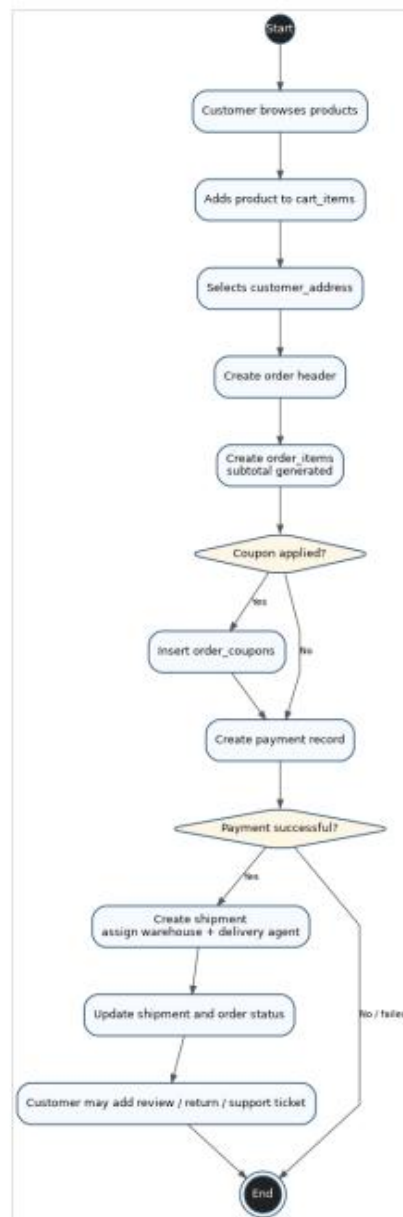
❖ Data Flow Diagram - Level 1

This Level 1 DFD breaks the RetailHub system into smaller processes. It is used to show how customer data, catalog data, orders, payments, fulfillment, and post-sales support move through the database. Each process writes to or reads from related table groups. The final analytics layer uses all these data stores to answer business questions on revenue, churn, stock, delivery, and support.



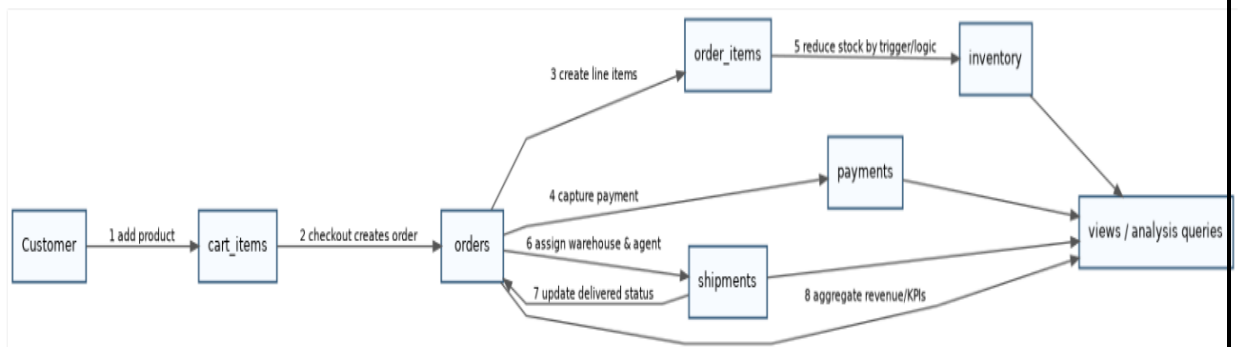
❖ Activity Diagram - Customer Order Flow

The activity diagram explains the step-by-step order journey from browsing to delivery. It is used to show process logic clearly, including decision points like coupon application and payment success. The diagram connects front-end user actions with backend database tables such as `cart_items`, `orders`, `order_items`, `payments`, `shipments`, `reviews`, `returns`, and `support_tickets`. It is useful for explaining project workflow in simple interview language.



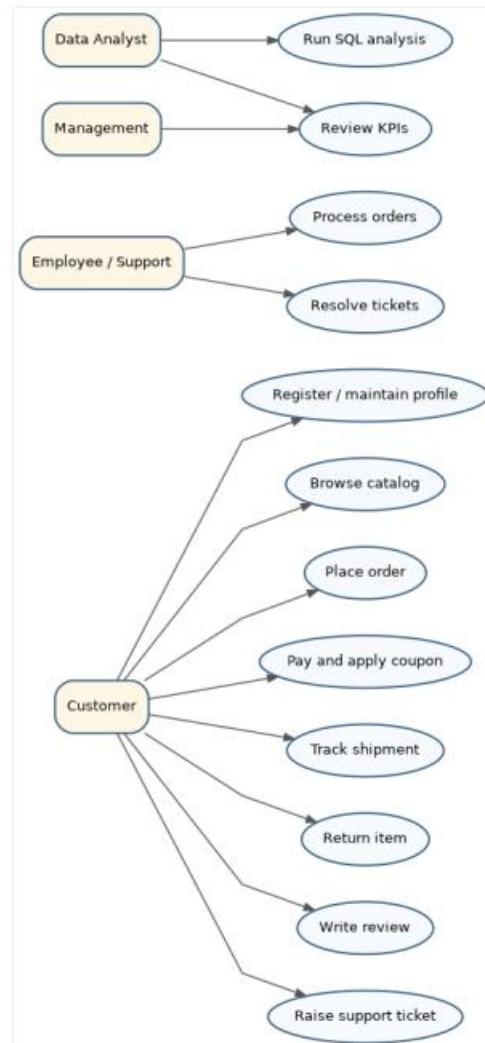
❖ Sequence Diagram - Order Lifecycle

The sequence diagram shows the order in which different database components are touched during checkout and fulfillment. It is used to explain timing: cart first, then order, then order items, payment, inventory update, shipment, and reporting. This diagram helps interviewers understand how one business event affects multiple tables. It also shows why foreign keys and transaction logic are important in an e-commerce database.



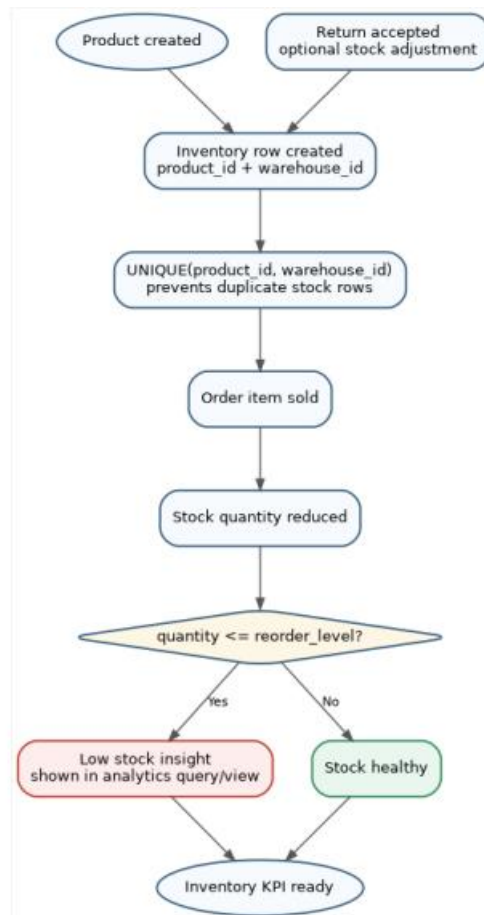
❖ Use Case Diagram

The use case diagram shows what each actor can do in the RetailHub system. It is used to explain project scope from a user requirement perspective. Customers can browse, order, pay, track, return, review, and raise tickets, while employees process orders and resolve support tickets. Analysts and management use the same database for reports and KPI review.



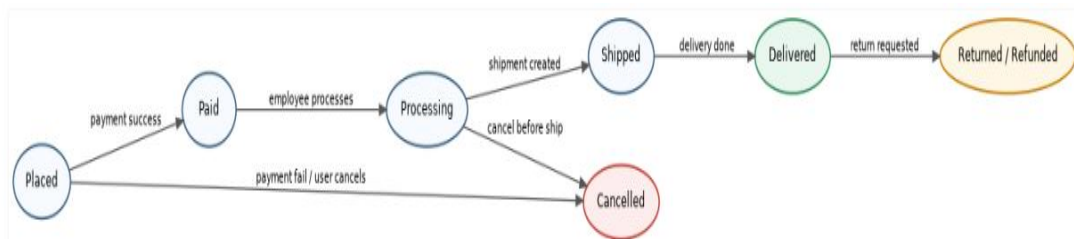
❖ Inventory Stock Flow Diagram

This diagram explains how product stock is stored and monitored across warehouses. It is used to show inventory logic in a way that is easy to discuss in interviews. The inventory table connects products and warehouses, and the unique product-warehouse rule prevents duplicate stock records. The low-stock decision point supports inventory health analysis and reorder planning.



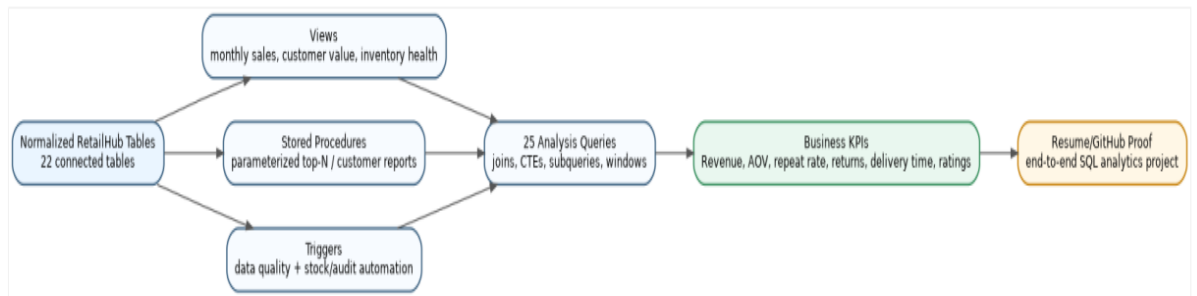
❖ Order Status State Diagram

This state diagram shows how an order can move from placed to paid, processing, shipped, delivered, cancelled, or returned. It is used to explain status-based business logic and reporting. Not every order reaches delivered status because payment failure or cancellation can stop the flow. Delivered orders can still move into return/refund analysis, which is important for measuring product quality and customer experience.



❖ Views, Procedures, Triggers and Analytics Diagram

This diagram explains the advanced SQL layer of the project. It is used to show that the project includes more than basic tables and joins. Views make repeated reporting easier, stored procedures allow reusable parameter-based analysis, and triggers support automation or data integrity. These components feed the final business KPIs used in the portfolio and resume.



❖ Keys and Constraints Diagram

This diagram explains why database keys and constraints are important. It is used to defend the schema design in an interview because clean analytics depends on clean relational data. Primary keys identify rows, foreign keys connect tables, composite keys handle many-to-many relationships, and CHECK/UNIQUE rules prevent invalid or duplicate data. These rules reduce confusion and make query results trustworthy.

